The Mathematics of Shocks via the Burgers Equation

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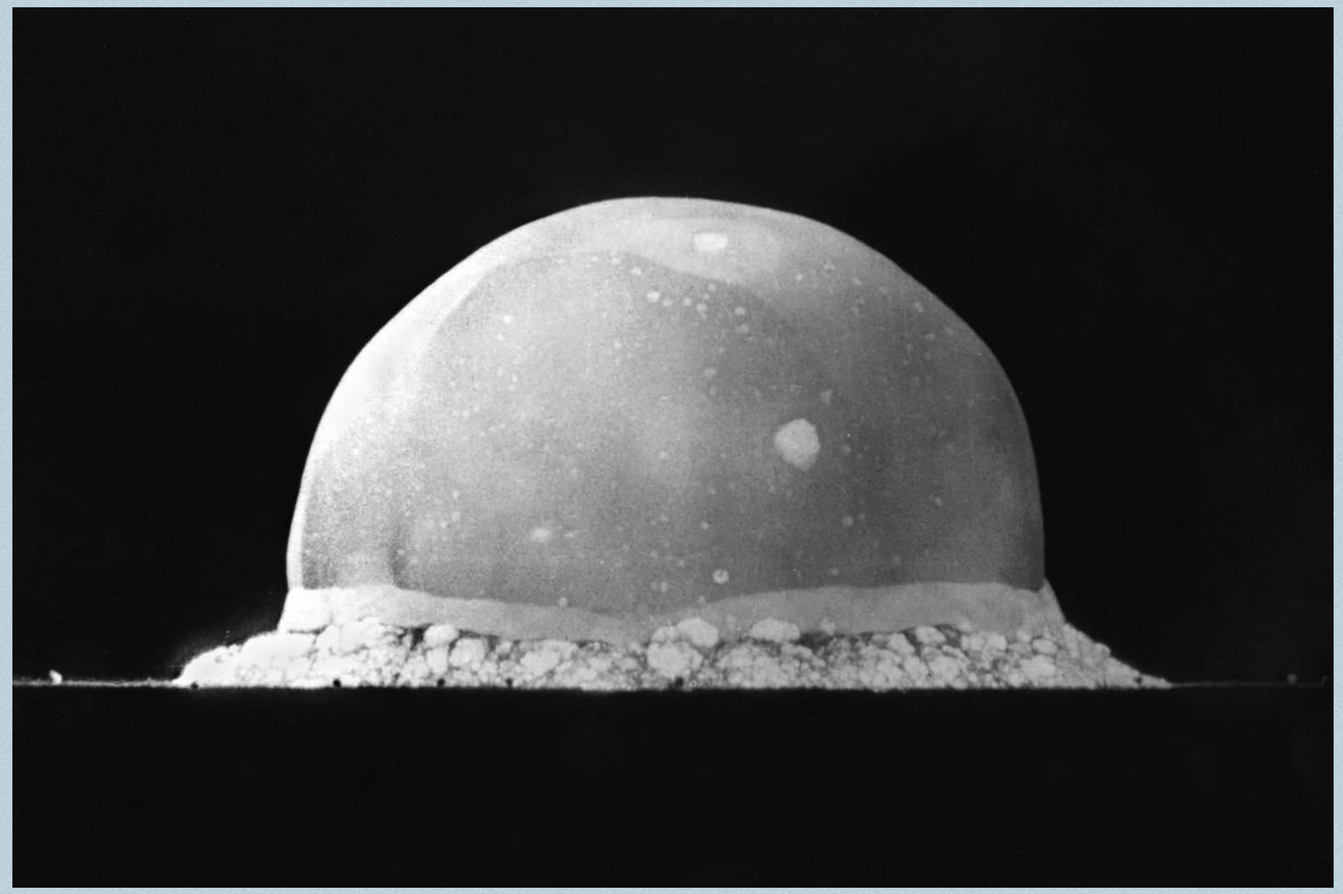
From Viscous to Inviscid: the Math



Silver plate photo of the Trinity test explosion.
A true example of a shock in fluid dynamics.

- The Euler equations have all the essential nonlinearity of the Navier Stokes equations, even though the stress is assumed to be isotropic.

- At the same time many naturally occurring fluids like air and water have very low values of viscosity.

- We can thus ask how one can pass from the viscous Navier Stokes equations to the inviscid Euler equations?

- This is a classical topic in fluid mechanics, and one that requires quite a bit of mathematical build up since the inviscid equations allow for solutions with a discontinuity (shocks) that strictly speaking break the PDEs.

- In this slide deck I use a classical model equation, or a PDE in one spatial variable and time, to show the mathematics of the limit of viscosity going to zero and why shocks don't "break" the physics.

$$A_t + AA_x = \nu A_{xx} \text{ Burgers Equation}$$

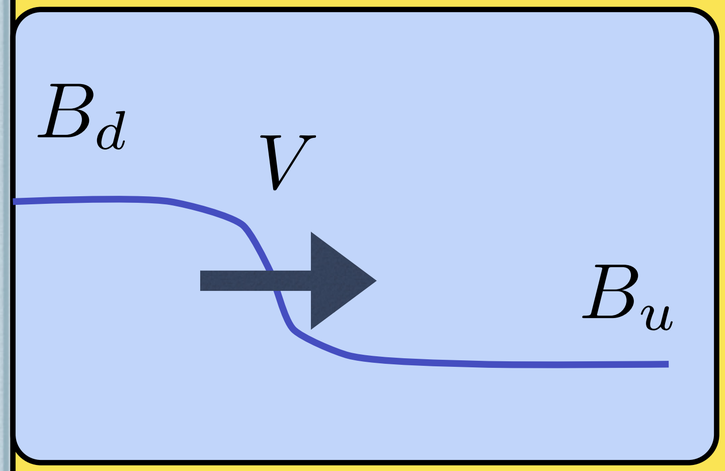
$$\xi = x - Vt \text{ travelling variable}$$

$$A(x,t) = B(\xi) \text{ travelling wave solution}$$

$$-VB' + \frac{1}{2}(B^2)' - \nu B'' = 0$$

$$-VB + \frac{1}{2}B^2 - \nu B' = C_1 \text{ integrating once}$$

- The Burgers equation allows us to understand the mathematics of the gas dynamics setting in a simpler context (one equation).
- First we will keep the second derivative term, then consider what happens when we drop it.
- A basic strategy for a nonlinear equation is to look for travelling solutions (notice these can't be plane waves since Burgers equation is nonlinear).
- We want to find travelling waves that are sharp transitions between two essentially constant regions.



$$-VB + \frac{1}{2}B^2 - \nu B' = C_1$$

Need to find V and C_1

$$B \to B_u, B_d \text{ as } \xi \to \pm\infty$$

$$B' \to 0 \text{ as } \xi \to \pm\infty$$

$$V = \frac{1}{2}(B_u + B_d)$$

$$C_1 = -\frac{1}{2}B_u B_d$$

- We can use the upstream (or in front of shock) and downstream (or behind shock) state to integrate once and to determine the speed of the travelling disturbance and the constant of integration.

- To integrate a second time we factor and use partial fractions.

$$(B - B_u)(B_d - B) = -2\nu B' \quad \text{factoring}$$

$$\frac{\xi}{\nu} = \frac{2}{B_d - B_u} \log \frac{B_d - B}{B - B_u} \quad \text{integrating}$$

$$\Delta B = B_d - B_u \quad \text{define the difference}$$

$$B(\xi) = B_u + \frac{\Delta B}{1 + \exp[\frac{\Delta B}{\nu}\xi]} \quad \text{final solution}$$

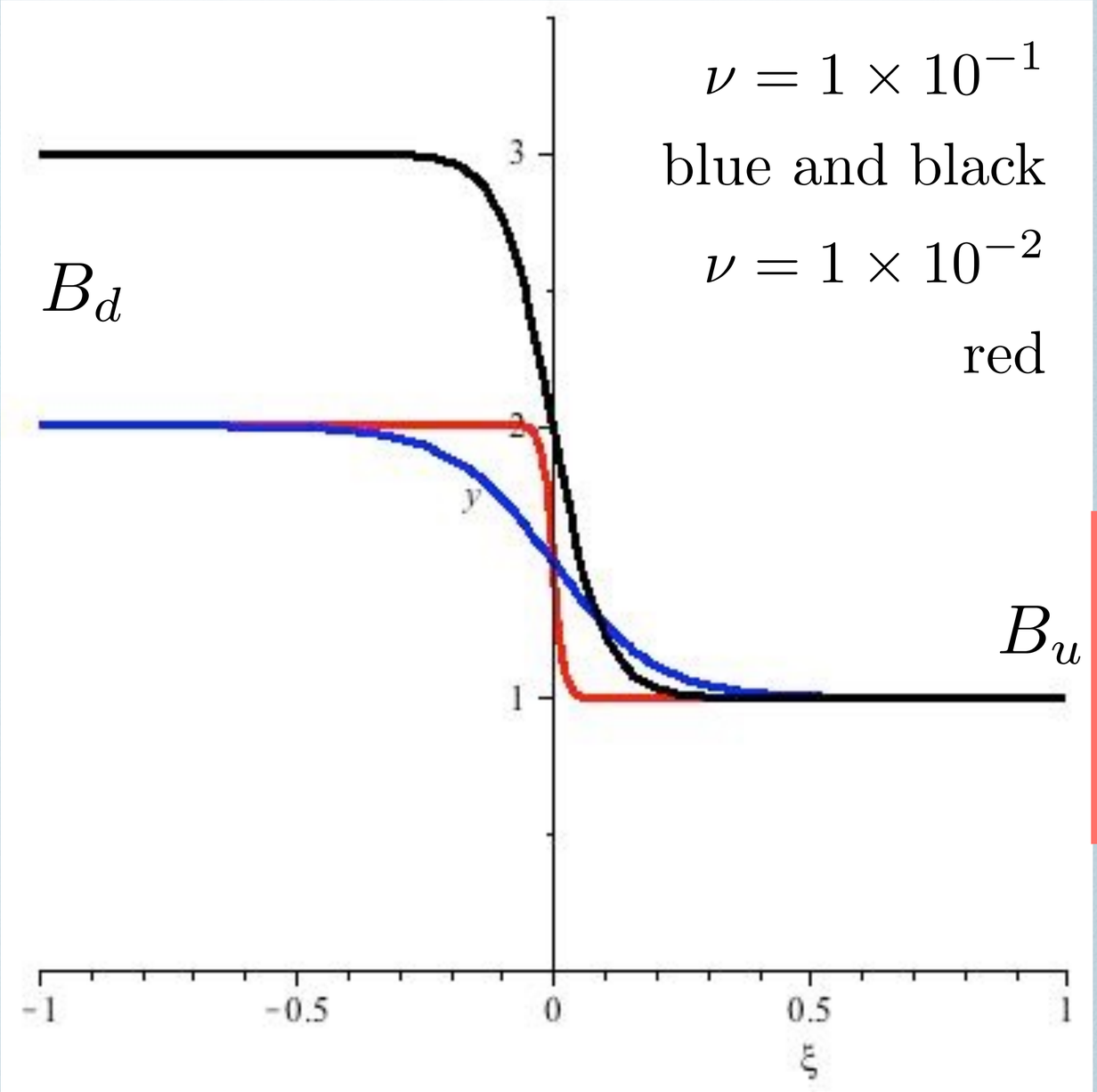
$$V = \frac{1}{2}(B_u + B_d)$$

- The first key point to notice is that the velocity of the travelling wave does NOT depend on the viscosity.

- The second key point to note is that the width of the wave depends inversely on viscosity, so lower viscosity means thinner waves.

- The third key point to note is that that thickness of the wave depends on the wave amplitude with larger waves being 'thinner'.

down stream or behind shock



upstream or ahead of shock

- We next want to figure out what happens when the viscosity tends to zero.
- This is the same sort of limit that gives the ideal gas equations.
- From the Burgers equation solution we see that the wave width goes to zero and thus the travelling wave solution tends to a discontinuity called a **shock.**
- We want to use some PDE theory to understand these solutions.
- The simplest problem is called the Riemann problem.

$$A_t + AA_x = 0 \text{ inviscid Burgers equation}$$
$$A(x,0) = A_d > A_u \text{ when } x < 0$$
$$A(x,0) = A_u > 0 \text{ when } x > 0$$

- We could define a characteristic curve in analogy with the xi variable in the travelling wave solution for the viscous Burgers equation:

$$\frac{dx}{dt} = A(x,t)$$

$$\frac{dx}{dt} = A(x,t)$$

- But for the Riemann problem on the last slide we have that

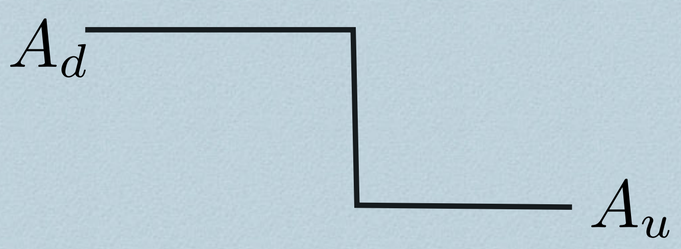
$$\frac{dx}{dt} = A_d \text{ when } x < 0$$

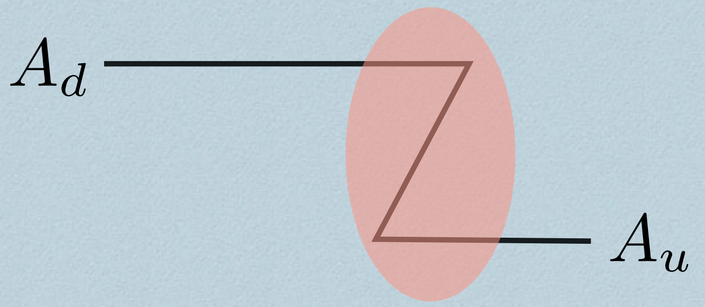
$$\frac{dx}{dt} = A_u \text{ when } x > 0$$

$$\text{But } A_d > A_u \text{ so the curves cross}$$

- Another way to see this is to write down an implicit solution using D'Alembert's solution for the advection equation (F(.) is given by the initial condition):

$$A(x,t) = F[x - A(x,t)t]$$





- So the implicit solution loses uniqueness immediately!

- How can we fix things? Let's return to the Burgers equation with viscosity.

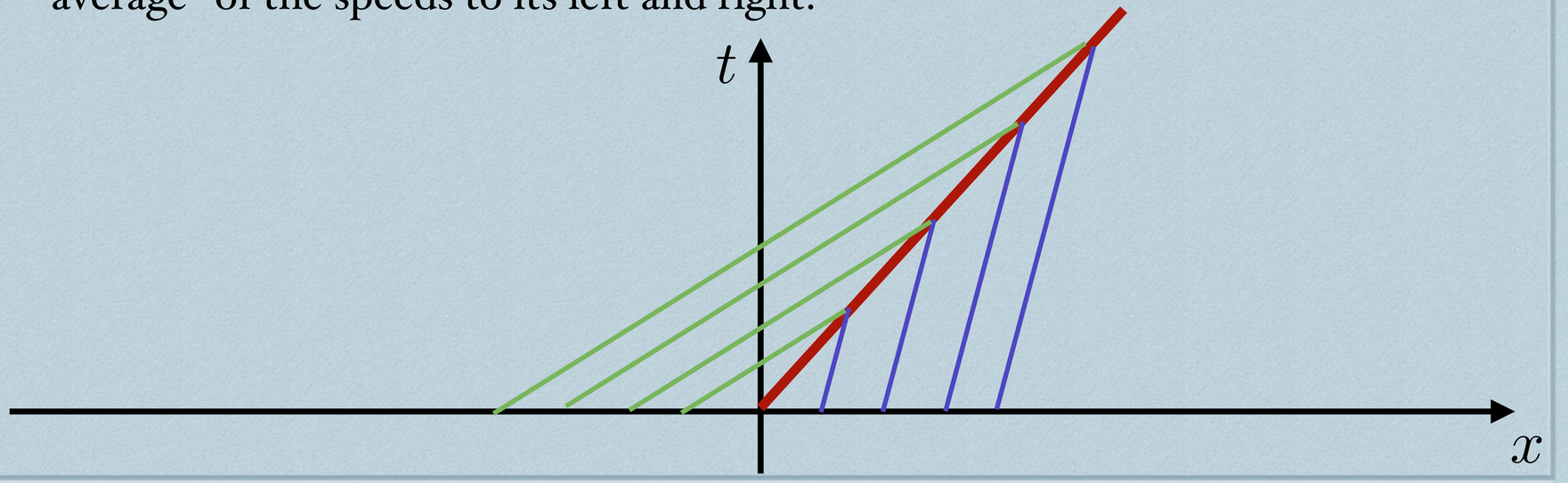
- We found for this case that the wave speed did not depend on viscosity so take the wave speed to match the viscous case and solve the initial value problem as:

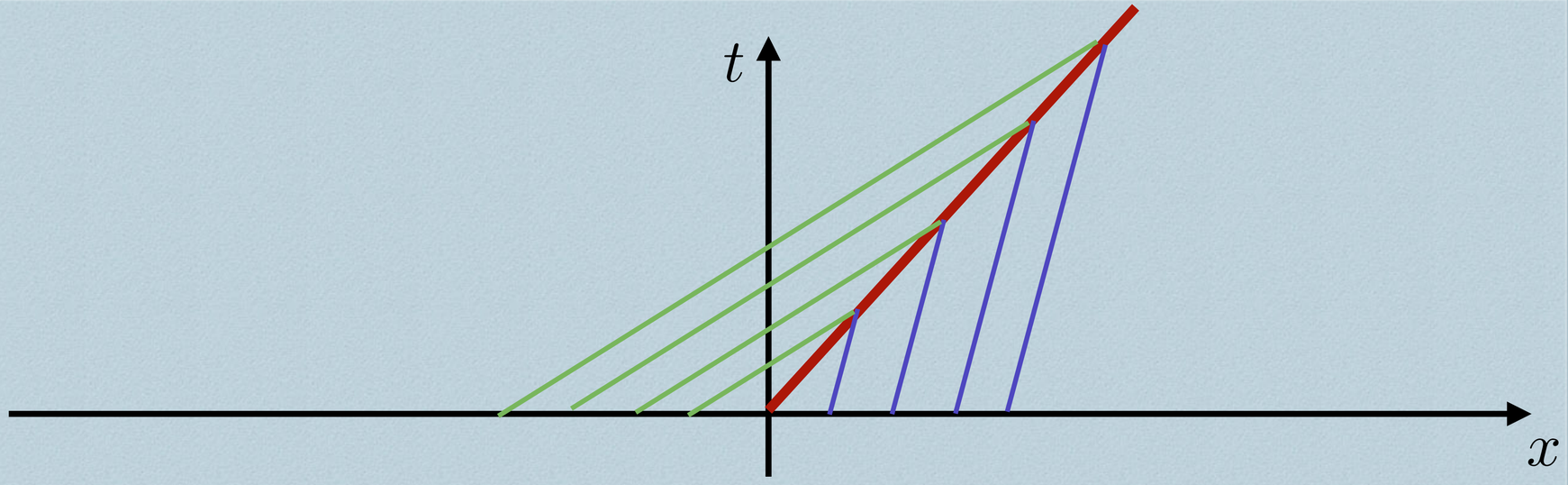
$$A(x,t) = A_d + (A_u - A_d)H\left(x - \frac{A_u + A_d}{2}t\right)$$

- Where H(.) is the Heaviside step function.

- All that remains to do is to fix the characteristic curves.

- To do this start with the discontinuity. At t=0 it is at x=0 and moves with the "average" of the speeds to its left and right.





- The steeper characteristics that start from the positive x axis are shown in blue.
- The less steep characteristics that start from the negative x axis are shown in green.
- The path the shock follows is shown in red.
- Returning to the original PDE you might notice that we have gone slightly out of bounds from a mathematical point of view.
- This is because the shock solution is not differentiable (it isn't even continuous!).

- To make things mathematically correct we need to slightly weaken our notion of what a solution is.

- The traditional solution is a formula which, when we substitute it into the equation gives the relation 0=0.

- For this to be true the solution must be differentiable as many times as the order of the PDE.

- For our case the solution is smooth away from the shock, so we seem to be left with the task of somehow "smudging" the single point at which the discontinuity happens.

- Of course this is exactly what viscosity does in the viscous Burgers equation which we were able to solve exactly for "shock-like" waves.

- Mathematicians call this type of solution a "weak" solution and weak solutions form a theoretical topic of study in PDE theory as well as a foundation for many modern numerical techniques like finite element methods.

$$\int_{-\infty}^{\infty} \phi(x)\,(A_t + AA_x)\,dx = 0$$

Where ϕ is an arbitrary, differentiable function

- You will recall that the integral "overlooks" values that occur at a single point.
- The above "weak formulation" of the PDE allows us to generalize the classical solution to situations like the shock.
- If A(x,t) is differentiable than the weak formulation is equivalent to the classical solution (for analysis fans, you can prove this).
- Numerical methods that handle shocks begin with the weak formulation and build approximation schemes based on it.
- The function ϕ is called a test function.

- There is one more piece of standard applied math notation that comes from gas dynamics and that is the notion of **Conservation Form**

$$A_t + AA_x = 0 \text{ usual inviscid Burgers eqn}$$

$$A_t + \left(\frac{1}{2}A^2\right)_x = 0 \text{ Conservation Form}$$

$$Q = \frac{1}{2}A^2 \text{ Definition of Q}$$

$$\frac{\partial}{\partial t}\int_{x_L}^{x_R} A(x,t), dx = -\left[Q(x_R) - Q(x_L)\right]$$

Matrix

- In 3D **Conservation Form** is an equation of the form:

$$\frac{\partial \vec{m}}{\partial t} + \nabla \cdot \mathbf{Q}(\vec{m}) = 0$$

- Integrating would now involve using Gauss' theorem on the divergence to get the flux of m through the boundary.

- There is another neat bit of mathematics that we can learn from the Burgers equation.
- Unlike linear PDEs, nonlinear PDEs have no general approach to their analytical solution. So what pre-computing practitioners sought to do was to transform these nonlinear PDEs to linear ones.
- The natural equation one would like to convert the Burgers equation to is a diffusion equation, since that retains the diffusive term.
- But getting rid of the nonlinear term is not easy.
- The answer is found in many texts and is called the Cole-Hopf transformation.
- The interesting question is how someone might have derived it.

$A_t + AA_x = \nu A_{xx}$ can be rewritten as $A_t = \left(-\frac{1}{2}A^2 + \nu A_x\right)_x$

This means a vector field $(A, -\frac{1}{2}A^2 + \nu A_x)$ is conservative.

This means there is a function v that satsifies $v_x = A$ and $v_t = -\frac{1}{2}A^2 + \nu A_x$; a bit of work gives $v_t = -\frac{1}{2}v_x^2 + \nu v_{xx}$.

Next let $v = f(\phi)$ to get $f'(\phi)\phi_t = -f'(\phi)^2\phi_x^2/2 + \nu(f'(\phi)\phi_x)_x$

$f'(\phi)\phi_t = -f'(\phi)^2\phi_x^2/2 + \nu f''(\phi)(\phi_x)^2 + \nu f'(\phi)\phi_{xx}$.

Now write as a diffusion equation on the left hand side to get $f'(\phi)\left(\phi_t - \nu\phi_{xx}\right) = \left[-f'(\phi)^2/2 + \nu f''(\phi)\right](\phi_x)^2$

We would like the term in the square brackets on the RHS to vanish.

Choosing $f(\phi) = -2\nu \ln \phi$ ensures this is the case.

- So we started with $A_t + AA_x = \nu A_{xx}$ and the helper definition $v_x = A$.
- The algebra led us to $v = -2\nu \ln \phi$ where ϕ satisfies $\left(\phi_t - \nu\phi_{xx}\right)/\phi = 0$.
- This means A satisfies $\dfrac{\partial}{\partial x}\left(\dfrac{\phi_t - \nu\phi_{xx}}{\phi}\right) = 0$
- Integrating with respect to x gives $\phi_t - \nu\phi_{xx} = \phi f(t)$ for some arbitrary function of time $f(t)$.
- The final trick is to write $f(t) = \dfrac{dg}{dt}$ since it's an arbitrary function anyway.
- Then let $\phi \to \exp(g)\phi$ and finally get a heat equation.
- To modern eyes this may seem a bit nuts.
- But if you consider pre-computer days and the whole real line then you get the following

- The IVP for $\phi_t - \nu\phi_{xx} = 0$ is given by $\phi(x,t) = \dfrac{1}{4\pi\nu t}\displaystyle\int_{-\infty}^{\infty} \phi(x',0)\exp\left(-\frac{(x-x')^2}{4\nu t}\right)dx'$

- The only question is how $\phi(x,0)$ is related to $A(x,0)$.

- Following the transformation through gives $\ln\phi(x,0) = -\dfrac{1}{2\nu}\displaystyle\int_0^x A(x',0)dx'$

- While not super helpful, practically, the Cole-Hopf transformation is a neat example of certain nonlinear PDEs being mappable to linear a PDE by a well defined set of operations that are mathematically well-understood.

- It isn't clear how wise it is to pursue something like Cole-Hopf in the age of readily available computation, but it does explain some of the language out there.

- This comes up again for soliton equations like the KdV in the theory of inverse scattering.

- We leave this for future work.

- One thing that is worth briefly surveying is where the Burgers equation comes from.

- It was studied by J. Burgers in the 1940s, but was previously considered by H. Bateman in the 1910s. This is often the way with science, and suggests naming equations after people is a pretty terrible idea.

- The wiki article has references and more: https://en.wikipedia.org/wiki/Burgers%27_equation

- As a fluids problem the Burgers equation is deceptive. It looks like a 1D momentum equation but is missing the pressure term (so there is nothing to drive the motion).

- It also misses the critical aspect of 3D fluid mechanics, namely vorticity stretching and tilting which are the dominant mechanisms by which turbulence sustains itself.

- As a numerics example the Burgers equation is undoubtedly useful, but misses a widely observed mechanism in natural, namely wave dispersion.